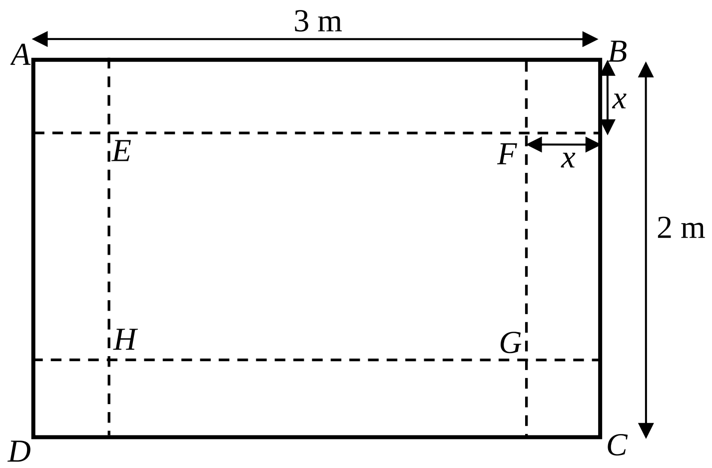
**Section A: Pure Mathematics [35 marks]**

**1** Find, algebraically, the set of values of $m$ for which the curves $y = x^2 - 2mx - 4.5$ and $y = 2x^2 - 3x + m$ do not intersect. [4]

**2**




The diagram shows a rectangular piece of metal sheet of length 3 m and breadth 2 m. A square of side $x$ m is removed from each corner of $ABCD$. The remaining shape is now folded along $EF$, $FG$, $GH$ and $HE$ to form an open rectangular box of height $x$ m.

**(i)** Show that the volume, $V$ m$^3$, of the box is given by $V = 4x^3 - 10x^2 + 6x$. [2]

**(ii)** Without using a calculator, find in surd form the values of $x$ that give stationary values of $V$. Determine the nature of these stationary values. [4]

**(iii)** Find the set of values of $x$ for which $V$ is decreasing as $x$ varies. [1]

**3** The curve $C$ has equation $y = 0.6(x - 2)^2 - 0.4^x$.

**(i)** Find the equation of the tangent to $C$ at the point $P$ where $x = 3$. Give your answer in the form $y = mx + c$, with $m$ and $c$ correct to 4 decimal places. [3]

**(ii)** This tangent meets the $y$-axis at point $A$ and the line $y = 2x$ at point $B$. If the ratio $PA : AB = 1 : n$, using similar triangles or otherwise, find $n$ to 4 decimal places. [4]

**4** **(a)** Differentiate $\ln\left(\dfrac{e^{2\sqrt{x}}}{\left(5x^4-1\right)^3}\right)$ with respect to $x$. [3]

**(b)** Show that $\dfrac{3}{x-3}-\dfrac{4}{2x+1}=\dfrac{2x+15}{(x-3)(2x+1)}$. Hence find $\displaystyle\int \dfrac{4x+30}{(x-3)(2x+1)}\,\mathrm{d}x$. [3]

**(c)** Given that $y = e^{1-2x^2}$. Find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$. Hence find $\displaystyle\int 8x\,e^{-2x^2+1} + \dfrac{1}{\sqrt[3]{x}}\,\mathrm{d}x$. [2]

**5** **(i)** Sketch the graph of $y = \dfrac{7x-6}{x-2}$, stating the equations of any asymptotes and the coordinates of any points of intersection with the axes. [3]

**(ii)** Find the exact $x$-coordinates of the points of intersection of $y=\dfrac{7x-6}{x-2}$ and the line $2y = 14x + 11$. Hence solve the inequality $2\left(\dfrac{7x-6}{x-2}\right) \le 14x + 11$. [3]

**(iii)** Write down as integral an expression for the area of the region bounded by $y=\dfrac{7x-6}{x-2}$ and $2y = 14x + 11$ and the line $x = K$ for $K \geq 3$. If the area is 10, find the value of $K$ to 3 decimal places. [3]

## Section B: Statistics [60 marks]

**6** At a secondary school Open House, the Principal wishes to find out how many primary school students would like to join her school.

**(i)** Explain how a systematic sample could be obtained from 5% of the primary school students. [2]

**(ii)** Give an advantage of using systematic sampling in this context. [1]

**(iii)** Give a disadvantage of using stratified sampling in this context. [1]

**7** A player throws three balls at a target. The probability that the player is successful in hitting the target with his first throw is $\dfrac{2}{5}$. Subsequently, the probability of hitting the target is $\dfrac{2}{3}$ if the preceding throw is successful and $\dfrac{1}{4}$ if the preceding throw is unsuccessful.

Find the probability that

**(i)** the second throw is successful, [2]

**(ii)** two out of three throws are successful, [2]

**(iii)** the second throw is successful given that two out of the three throws are successful. [2]

**8** **(a)** A group of 200 men and women registered for an advanced cooking course in a culinary school. The school tested the group and found that 10 men and 15 women have an extraordinary sense of taste and smell. A person is randomly selected from the group.

The events $E$ and $W$ are defined as follows.

$E$: the person selected has extraordinary sense of taste and smell.

$W$: the person selected is a woman.

Find

**(i)** $\mathrm{P}(E)$, [1]

**(ii)** $\mathrm{P}(E \cap W)$, [1]

**(iii)** the number of women in the group if $\mathrm{P}(E \cup W) = 0.47$. [2]

**(b)** Events $K$ and $L$ are such that $\mathrm{P}(L) = 0.3$, $\mathrm{P}(K \mid L) = 0.8$ and $\mathrm{P}(L \mid K) = 0.6$.

Find $\mathrm{P}(K)$. [2]

Are events $K$ and $L$ independent? Justify your answer. [1]

**9** A warehouse contains a large number of blue, red and green mugs in the ratios 1 : 3 : 2 respectively.

A random sample of 20 mugs is selected from the warehouse.

The number of green mugs selected out of 20 is denoted by the random variable $G$.

State in the context of this question, an assumption needed to model $G$ by a binomial distribution. [1]

Find the probability that the sample of 20 mugs contains

**(i)** exactly 4 green mugs, [2]

**(ii)** at least 2 and less than 8 green mugs, [2]

**(iii)** more than 13 mugs that are either blue or red. [2]

Another random sample of 100 mugs is drawn. Using a suitable approximation, find the probability that the sample contains more than 20 and at most 50 green mugs. [3]

**10** A company claims that the mean mass of a 'Star' Chocolate bar is 300 grams. A consumer watchdog claims that the mean mass of a 'Star' Chocolate bar is less than 300 grams.

Denoting the mass of a 'Star' Chocolate bar by $x$ grams, the results of a random sample of 125 'Star' Chocolate bars, are summarised by

$$\sum(x-300) = -50,\ \sum(x-300)^2 = 851.$$

**(i)** Find the unbiased estimates of the population mean and variance. [2]

**(ii)** Test at the 5% significance level whether the claim by the consumer watchdog is valid. [4]

**(iii)** Explain, in the context of the question, the meaning of 'at the 5% significance level'. [1]

**(iv)** Explain whether it is necessary to assume a normal distribution for the test to be valid. [1]

**(v)** The company thinks that the claim made by the consumer watchdog is incorrect. Based on the results in part **(ii)**, explain if the company's claim is justified by testing against the alternative hypothesis of the mean is not 300 at 5% significance level. [2]

**11** A particular orchard supplies mangoes. The mass of a mango has mean 0.6 kg and standard deviation 0.05 kg.

In normal time, the orchard randomly packs the mangoes into 'large' plastic bag and 'small' plastic bag for sale. (Assume that the mass of plastic bags are negligible in this question.)

Each 'large' plastic bag contains 100 mangoes.

**(i)** Explain clearly why the distribution of the mass of a randomly chosen 'large' plastic bag of mangoes is approximately normal with mean 60 kg and standard deviation 0.5 kg. [1]

**(ii)** Find the probability that a randomly chosen large plastic bag of mangoes has a mass exceeding 61 kg. [1]

Mass of each 'small' plastic bag of mangoes has an approximately normal distribution with mean 30 kg and variance 0.125 kg$^2$.

**(iii)** Find the probability that the mass of five randomly chosen 'large' plastic bags of mangoes differs from ten times the mass of a randomly chosen 'small' plastic bag of mangoes by at least 1 kg. [3]

Each 'small' plastic bags and 'large' plastic bags of mangoes are sold at \$18 per kg and \$16 per kg respectively. David buys ten 'small' plastic bags of mangoes while Peter buys five 'large' plastic bags of mangoes.

**(iv)** Find the probability that David pays more than Peter by at least \$650. [3]

During a special offer, 36 mangoes are randomly packed into a 'special' plastic bag that cost \$$m$ per kg. Find the maximum value of $m$ such that the probability of a randomly chosen 'special' plastic bag of mangoes costing less than \$320, is more than 0.8. Give your answer to 2 decimal places. [3]

**12** Shoe size may be used in forensic science to estimate the height of an unknown person.

The table below shows the shoe sizes (inch) and heights (cm) of 9 males.

| Shoe Size, $x$ | 7.5 | 7.0 | 9.0 | 8.0 | 8.5 | 10.0 | 11.0 | 12.0 | 6.0 |
|---|---|---|---|---|---|---|---|---|---|
| Height, $y$ | 162 | 160 | 169 | 162 | 167 | 170 | 182 | 184 | 152 |

**(i)** Draw a scatter diagram for the data. Find $(\bar{x}, \bar{y})$ and mark on your scatter diagram where $(\bar{x}, \bar{y})$ is the mean point of the sample. [2]

**(ii)** Find the product moment correlation coefficient and comment on its value in the context of the question. [2]

**(iii)** Find the equation of the regression line of $y$ on $x$, and draw this line on your scatter diagram. [2]

**(iv)** Using the regression line from **(iii),** estimate the shoe size to the nearest half inch when the height of the man is 171 cm. Comment on the reliability of the estimate. [2]

**(v)** A new regression line of $y$ on $x$, $y = 5.07x + 123$ is obtained when another data pair was added. If the shoe size of this data pair is 6.5, find the corresponding height to the nearest cm. [3]

**(vi)** Susan found her great-grandfather's shoes in an old rusty iron box and its size is 5.5. She used the regression line in part **(v)** to predict his height. Comment on the reliability of this prediction. [1]